Classifying 5M bank documents on a ₹1,00,000 budget: how one document moves through the pipeline

How to read this flowchart

1. Follow one document from the top. Each blue box is a step that buys evidence about it; its cost is in brackets, and the steps get dearer as you go down.

2. The document carries a probability for each of the 15 classes. It starts from the class mix expected for its file format (never 0), and every blue step updates it (Bayes' rule).

3. After each step a diamond decides. **YES**: one class is sure enough, so stop and accept. **NO**: buy the next, dearer step. Orange diamonds use the movable gate τ; yellow ones are fixed checks. The purple box splits KYC IDs into Proof of Identity or Proof of Address.

4. Every document ends in one of two places: ACCEPT (green), with its label, confidence and the step that settled it, or a person (red).

5. The three boxes at the top right apply at every orange diamond: what τ is, the critical safety net, and the budget controller that moves τ.

Why 95%: an accepted document is then wrong at most 5% of the time, and each next step costs several times the last, so stopping as soon as one class is sure is the cheapest way to the bar (macro-F1 ≥ 0.85, critical recall ≥ 0.90).

Why τ moves only at the gates before the LLM: reading pages is where most of the money goes, so that is where the controller trades accuracy for spend; never for critical documents.

S3 bucket
5M objects + the free manifest
(key, size, checksum)

L0 Open the file (3 ms, no model)
true type from the bytes; trusted visible text vs scan;
emails and zips unpacked; start from P(class | format)

Same checksum as a file seen before? — **YES** → Copy the first copy's label (free)

NO

Trusted visible text on page 1? — **YES**: digital files → **L2 on the free text** (~1 ms)
letter-piece model + ID checks
(checksum-valid IDs are evidence, not verdicts)

NO: scans, photos

Top class ≥ τ? — **NO**: add layout

YES

Bundle splitter (multi-page files)
every page: L1 layout + page-number line;
new document where P(boundary) ≥ 0.40;
"Page 3" never starts one, "Page 1" always does

L1 Layout (0.11 s/page)
DiT (frozen, MIT) + 6 readable signals
+ logistic regression, temperature-calibrated
settles 64% of scanned documents at 100.0%

Top class ≥ τ? — **YES** → Is it a KYC ID document? — **YES** → **Address question**
full read of the card;
valid PIN + address label or state
→ PoA, otherwise PoI

NO

NO: buy the cheapest read that can split what's left

Reading ladder + L2 text model
rung 2 selective read: top and largest lines (0.48 s)
rung 3 full page (2.2 s)
rung 4 next informative page + last page
each rung replaces the last rung's text evidence
legal/property: later pages first (the e-stamp page is shared)

Top class ≥ τ? (after each rung) — **NO**: next rung

YES

NO: after rung 4

L3 LLM: Haiku via `claude -p`, tools off
masked text (IDs replaced by their type);
2-5 candidate classes + "none of these";
answer used as a sensor with a measured error rate
mode: quote, ~₹0.038 per call

P(KYC) × P(answer) ≥ 95%? — **NO** → Top class ≥ 95%? — **YES**

YES

NO: or the LLM says none of these

L4 Human review
no price quoted: counted, not costed
about 2,600 of 5M documents (dev-test 0.1%)

a person labels it

ACCEPT
label, confidence, the step that settled it, its cost
(bundles: one row per document inside, with its page range)

Apply at every orange diamond

Critical safety net
if P(Legal, Property, PoI or PoA) ≥ 10%,
the document's gate never drops below 95%

τ = the gate at every orange diamond:
95% by default; the controller moves it
between 85% and 99%

moves τ

Budget controller
after every batch: spend vs a straight plan line
target ₹50k, 10% reserve (aims at ₹45k)
over plan → lower τ
under plan → raise τ only if that is measured
to buy accuracy (it did not: held at 95%)

Measured (synthetic documents)
dev-test: macro-F1 0.998; critical recall 1.00 on all four
Lockbox A (sealed, 40% unseen layouts, run once): macro-F1 0.990
Lockbox C (fresh, run once): macro-F1 0.982; critical recall 1.00
reaching the LLM: 0.4%; reaching a person: 0.1%
5M cost: ₹21,601 (0.43 paise a document) vs ₹1,30,170 brute force